



Session 9B: 2D Geometry — Lines, Conics, Distance, and Shape

Phase 2 — Classical Techniques | 90 min

Lines form the alphabet, conics form the words, and distance binds them all. Pure algebra, systematic geometric reasoning — no calculus required.

Prerequisite: 9A1 (function fundamentals), 9A2 (graph drawing toolkit). Composition, inverse, even/odd symmetry, and transformations are assumed. If you need review, see 9A1 Parts E–J.

Part A: The Line — All Five Forms, All Properties

A line is the simplest geometric object. Master every way to write it and every way to measure it.

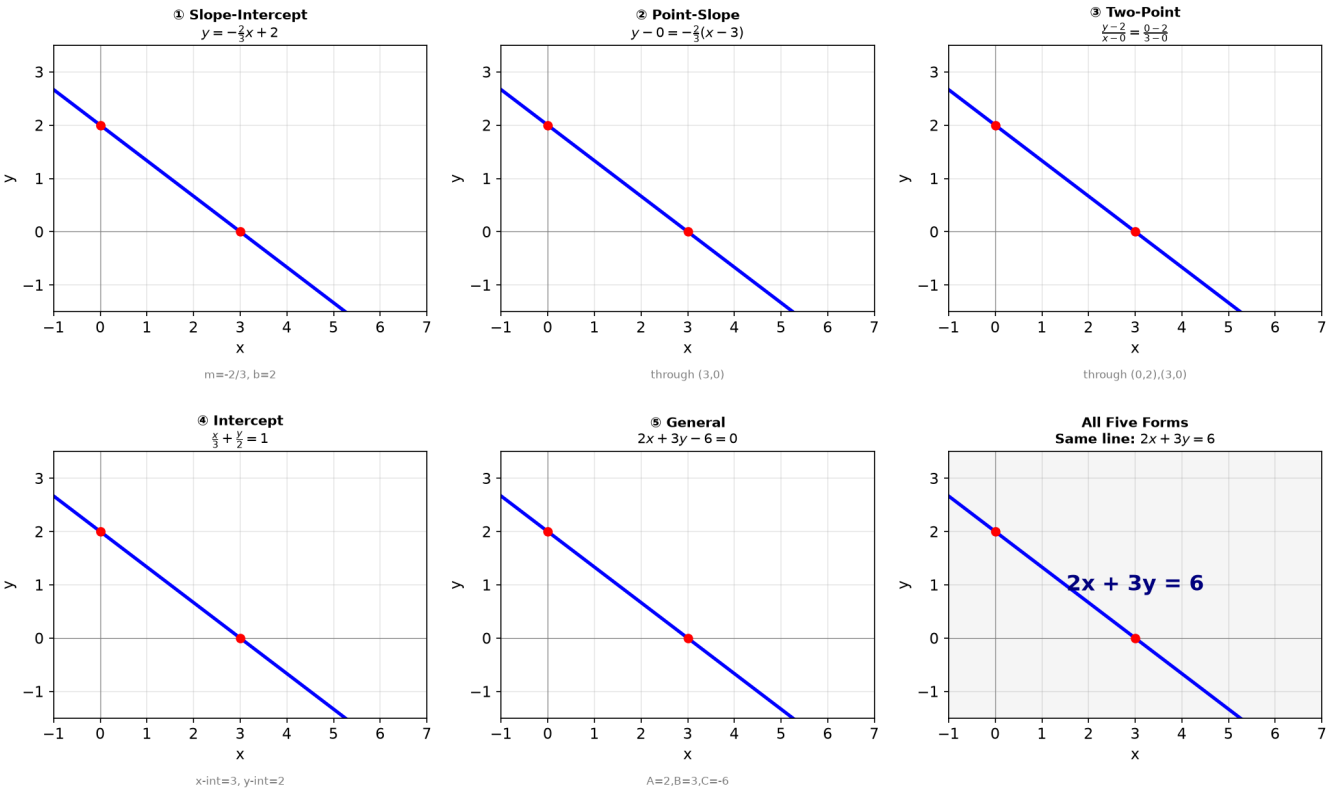
Example 1: The Five Forms of a Line — When to Use Each

A line through (x_1, y_1) with slope m . Five ways to write it — each suited to a different task.

#	Form	Equation	Best used for
①	Slope-intercept	$y = mx + b$	Quick graphing, reading slope and y -intercept
②	Point-slope	$y - y_1 = m(x - x_1)$	Building from one point + slope
③	Two-point	$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$	Building from two known points

#	Form	Equation	Best used for
④	Intercept	$\frac{x}{a} + \frac{y}{b} = 1$	When x - and y -intercepts are known
⑤	General	$Ax + By + C = 0$	Distance formulas, intersection, systems

The Five Forms of a Line



Graph 9B-A1: The same line $2x + 3y = 6$ expressed in all five forms. Each form highlights a different feature — slope, intercepts, points, or the normal vector.

Example 2: Converting Between Forms — A Single Line, Many Faces

Line through $(2, 1)$ with slope $m = \frac{3}{4}$.

① **Point-slope** (start here): $y - 1 = \frac{3}{4}(x - 2)$.

② **Slope-intercept** (solve for y): $y = \frac{3}{4}x - \frac{3}{2} + 1 = \frac{3}{4}x - \frac{1}{2}$.
 $\rightarrow m = \frac{3}{4}, b = -\frac{1}{2}$.

③ **General** (multiply by 4, move all to one side): $4y = 3x - 2 \rightarrow 3x - 4y - 2 = 0$.
 $\rightarrow A = 3, B = -4, C = -2$. Normal vector: $(3, -4)$.

④ **Intercept** (find where $x = 0$ and $y = 0$):

$$x = 0: y = -\frac{1}{2} \rightarrow b = -\frac{1}{2}.$$

$$y = 0: 0 = \frac{3}{4}x - \frac{1}{2} \rightarrow x = \frac{2}{3} \rightarrow a = \frac{2}{3}.$$

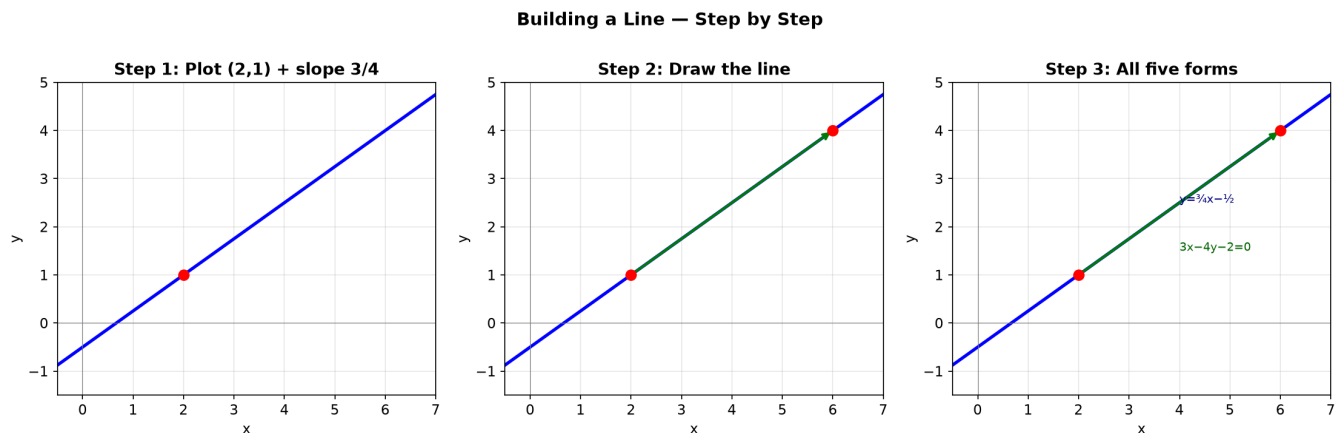
$$\text{Form: } \frac{x}{2/3} + \frac{y}{-1/2} = 1, \text{ or } \frac{3x}{2} - 2y = 1.$$

⑤ **Two-point** (pick any two points):

At $x = 2: y = 1 \rightarrow (2, 1)$. At $x = 6: y = 4 \rightarrow (6, 4)$.

$$\frac{y-1}{x-2} = \frac{4-1}{6-2} = \frac{3}{4}.$$

Every form contains the same line. Converting between them is a core skill — each form solves a different type of problem.



Graph 9B-S1: Building a line in three steps. Step 1 — Plot point $(2, 1)$ and use slope $3/4$ to find a second point. Step 2 — Draw the line. Step 3 — Label all five forms on the same line.

Example 3: Slope as a Rate — Rise Over Run, Angle, and Parallelism

Slope definition: $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$.

Slope from angle: $m = \tan \theta$, where θ is the angle the line makes with the positive x -axis.

- $m > 0$: line rises left to right ($0^\circ < \theta < 90^\circ$).
- $m < 0$: line falls left to right ($90^\circ < \theta < 180^\circ$).
- $m = 0$: horizontal ($\theta = 0^\circ$).
- m undefined: vertical ($\theta = 90^\circ$).

Parallel lines: $m_1 = m_2$ (same slope, same steepness).

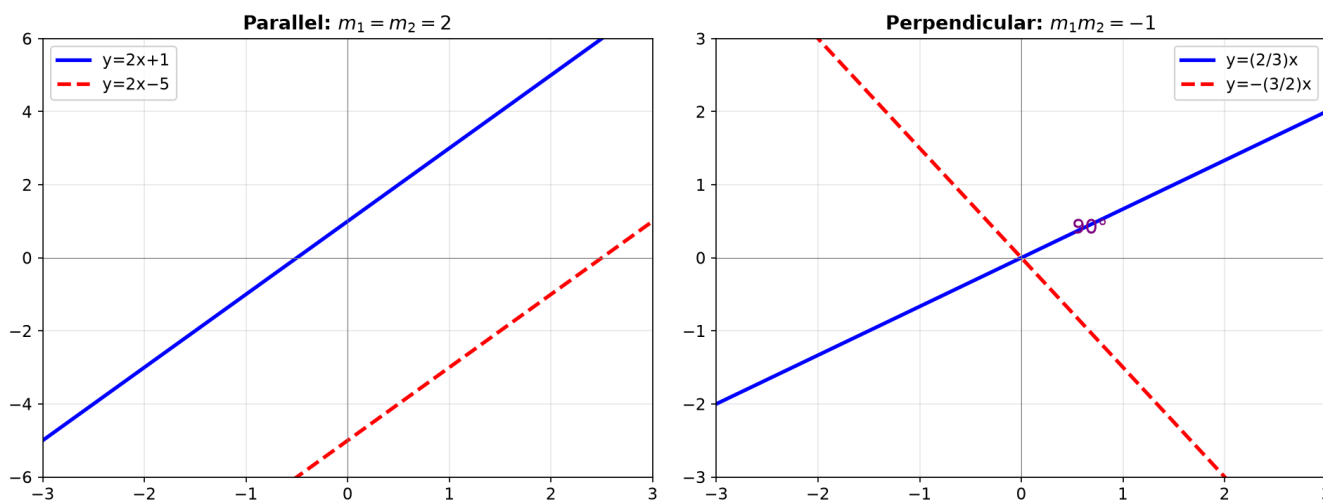
$y = 2x + 1$ and $y = 2x - 5$ are parallel — same slope 2, different intercepts.

Perpendicular lines: $m_1 \cdot m_2 = -1$ (slopes are negative reciprocals).

$y = \frac{2}{3}x$ is perpendicular to $y = -\frac{3}{2}x$, since $\frac{2}{3} \cdot \left(-\frac{3}{2}\right) = -1$.

Why $m_1 m_2 = -1$? Rotating a line by 90° sends $\Delta x \rightarrow -\Delta y$ and $\Delta y \rightarrow \Delta x$, so the new slope is $-\frac{1}{m}$.

Parallel and Perpendicular Lines



Graph 9B-A2: Left — Two parallel lines ($m = 2$), same steepness, different heights. Right — Two perpendicular lines ($m = 2/3$ and $m = -3/2$), the product of slopes equals -1 .

Example 4: Angle Between Two Lines

Given two lines with slopes m_1 and m_2 , the acute angle ϕ between them:

$$\tan \phi = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

Why this works: $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$. Set $\alpha = \tan^{-1}(m_2)$, $\beta = \tan^{-1}(m_1)$.

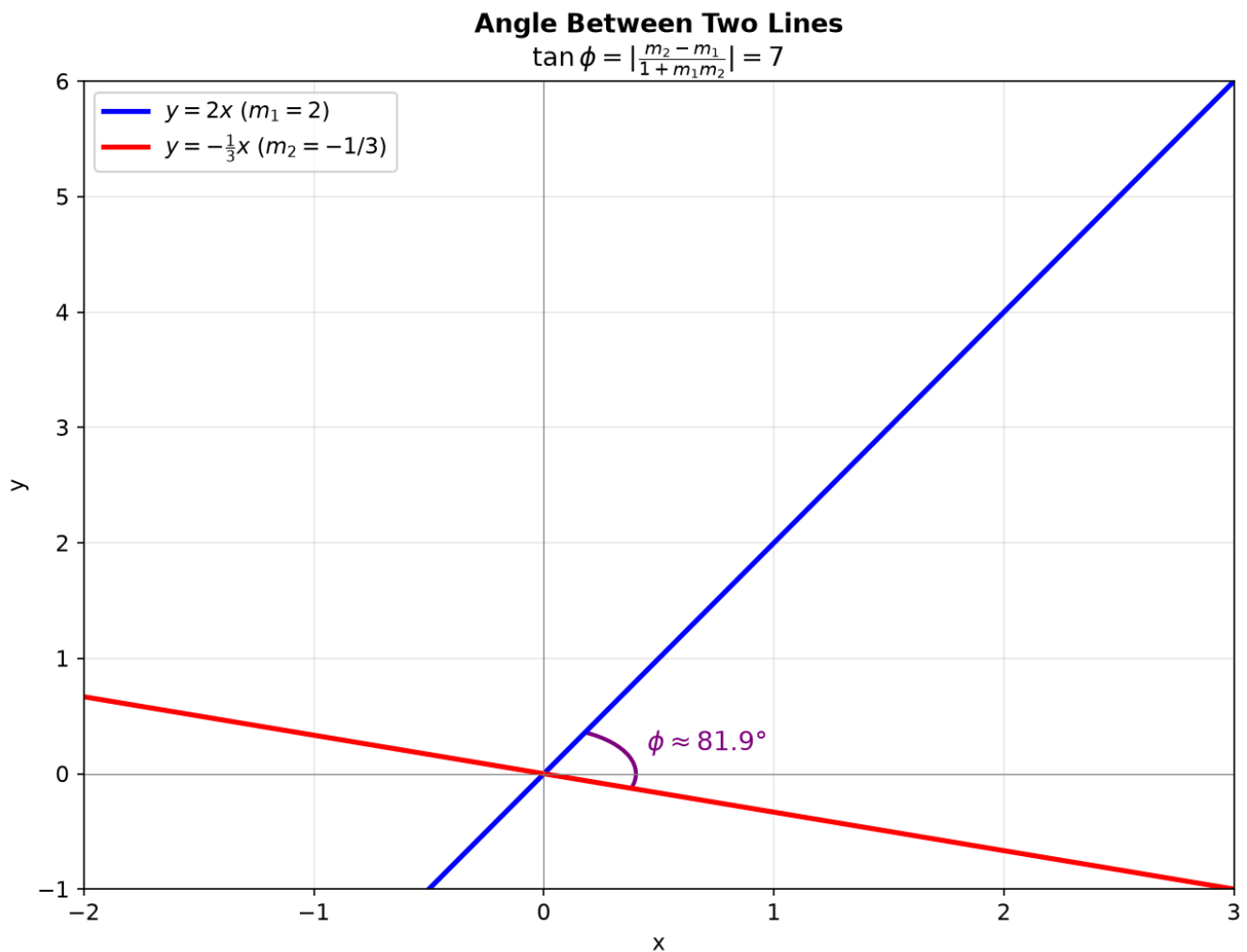
Example: Between $y = 2x$ ($m_1 = 2$) and $y = -\frac{1}{3}x$ ($m_2 = -\frac{1}{3}$):

$$\tan \phi = \left| \frac{-1/3 - 2}{1 + 2(-1/3)} \right| = \left| \frac{-7/3}{1 - 2/3} \right| = \left| \frac{-7/3}{1/3} \right| = 7.$$

$$\phi = \tan^{-1}(7) \approx 81.9^\circ.$$

Special cases:

- If $1 + m_1 m_2 = 0$ (denominator zero), the lines are **perpendicular** ($\phi = 90^\circ$).
- If $m_1 = m_2$ (numerator zero), the lines are **parallel** ($\phi = 0^\circ$).



Graph 9B-A3: The angle ϕ between lines $y = 2x$ and $y = -x/3$ is $\tan^{-1}(7) \approx 81.9^\circ$. The formula comes from the tangent subtraction identity.

Example 5: Midpoint and Section Formula — Dividing a Segment

Midpoint of (x_1, y_1) and (x_2, y_2) :

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$(2, 5)$ and $(8, -1) \rightarrow$ midpoint $(5, 2)$.

Section formula — point that divides the segment in ratio $m : n$ (from (x_1, y_1) to (x_2, y_2)):

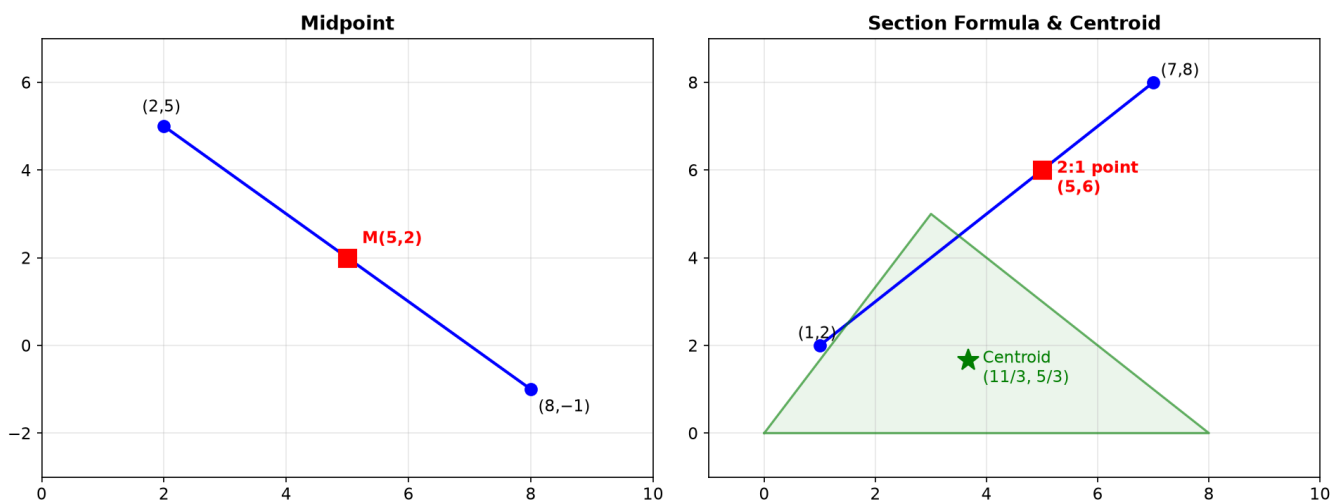
$$P = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

$(1, 2)$ to $(7, 8)$, ratio $2 : 1$ from the first point:

$$P = \left(\frac{2(7) + 1(1)}{3}, \frac{2(8) + 1(2)}{3} \right) = (5, 6).$$

Centroid of a triangle (average of vertices): $G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$.

Midpoint and Section Formula



Graph 9B-A4: Left — Midpoint bisects the segment. Right — Section formula divides the segment in ratio 2:1. The centroid of the triangle is the average of its three vertices.

Up to here (Part A) : Five line forms — slope-intercept, point-slope, two-point, intercept, general. Parallel ($m_1 = m_2$), perpendicular ($m_1 m_2 = -1$). Angle between lines. Midpoint and section formula.

Part B: Distance in 2D — Systematic Methods

Distance is geometry's fundamental measure. Two points, point-to-line, line-to-line, point-to-curve — each has a formula derived from a single principle: the shortest path is perpendicular.

Example 6: Point-to-Point — The Pythagorean Foundation

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

$$(3, 1) \text{ to } (7, 4): d = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = 5.$$

This is a 3-4-5 right triangle — the distance formula **is** the Pythagorean theorem.

Example 7: Point-to-Line — Deriving the Formula

Point (x_0, y_0) to line $Ax + By + C = 0$:

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$$

Where does this come from?

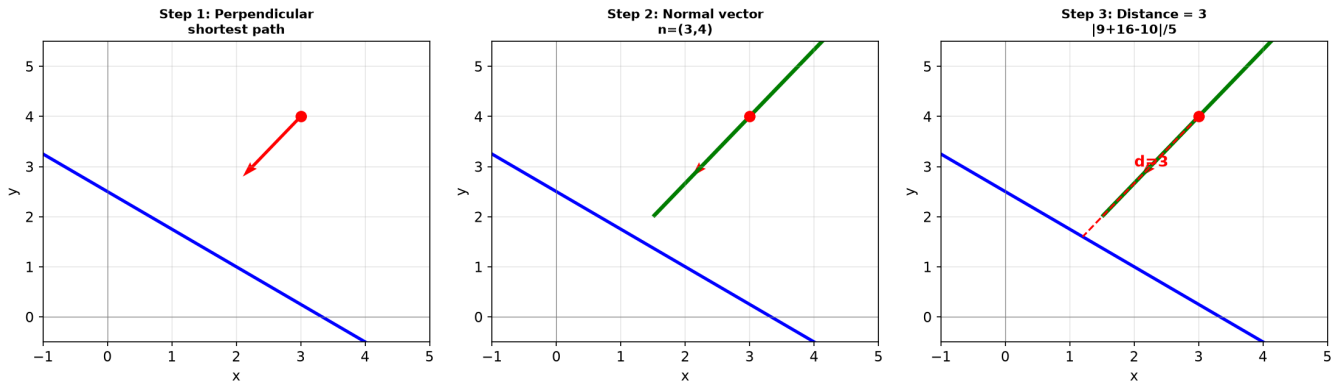
- ① The shortest segment from the point to the line is **perpendicular** to the line.
- ② The normal vector $\vec{n} = (A, B)$ points perpendicular to the line.
- ③ Project the vector from any point on the line to (x_0, y_0) onto the normal direction.
- ④ The length of this projection is $\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$.

Example: $(3, 4)$ to $3x + 4y - 10 = 0$.

$$d = \frac{|3(3) + 4(4) - 10|}{\sqrt{3^2 + 4^2}} = \frac{|9 + 16 - 10|}{5} = \frac{15}{5} = 3.$$

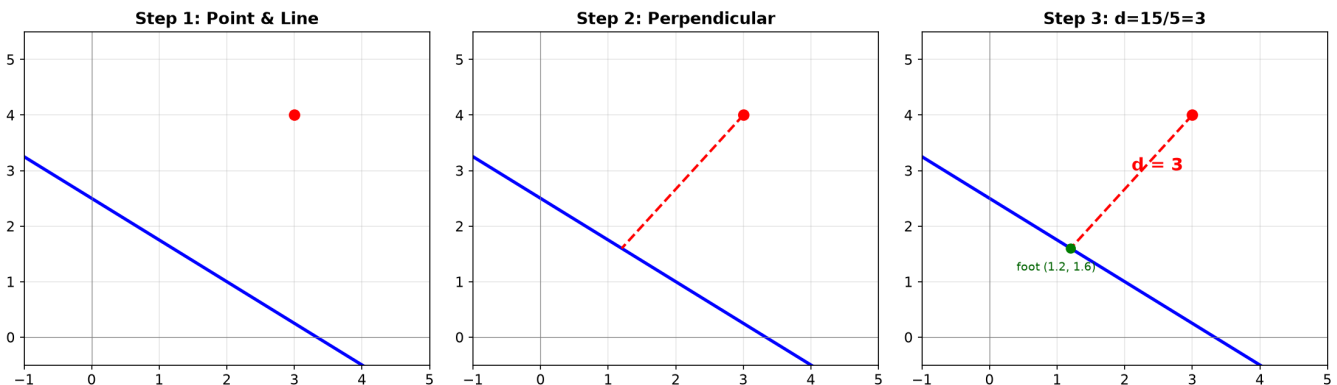
Numerator intuition: $Ax_0 + By_0 + C$ measures how "wrong" the point is — plug it into the line equation. If the point lies on the line, the expression equals 0 and $d = 0$.

Deriving Point-to-Line Distance



Graph 9B-B1: Deriving the point-to-line distance formula. Step 1 — The shortest path is perpendicular (red dashed). Step 2 — The normal vector (A, B) points perpendicular to the line. Step 3 — The distance is the projection length.

Point to Line Distance — Step by Step



Graph 9B-S2: Computing distance from $(3, 4)$ to $3x + 4y = 10$ in three steps. Step 1 — Plot point and line. Step 2 — Draw perpendicular segment. Step 3 — Apply formula: $|9 + 16 - 10|/5 = 3$. The foot of the perpendicular is at $(1.2, 1.6)$.

Example 8: Distance Between Two Parallel Lines

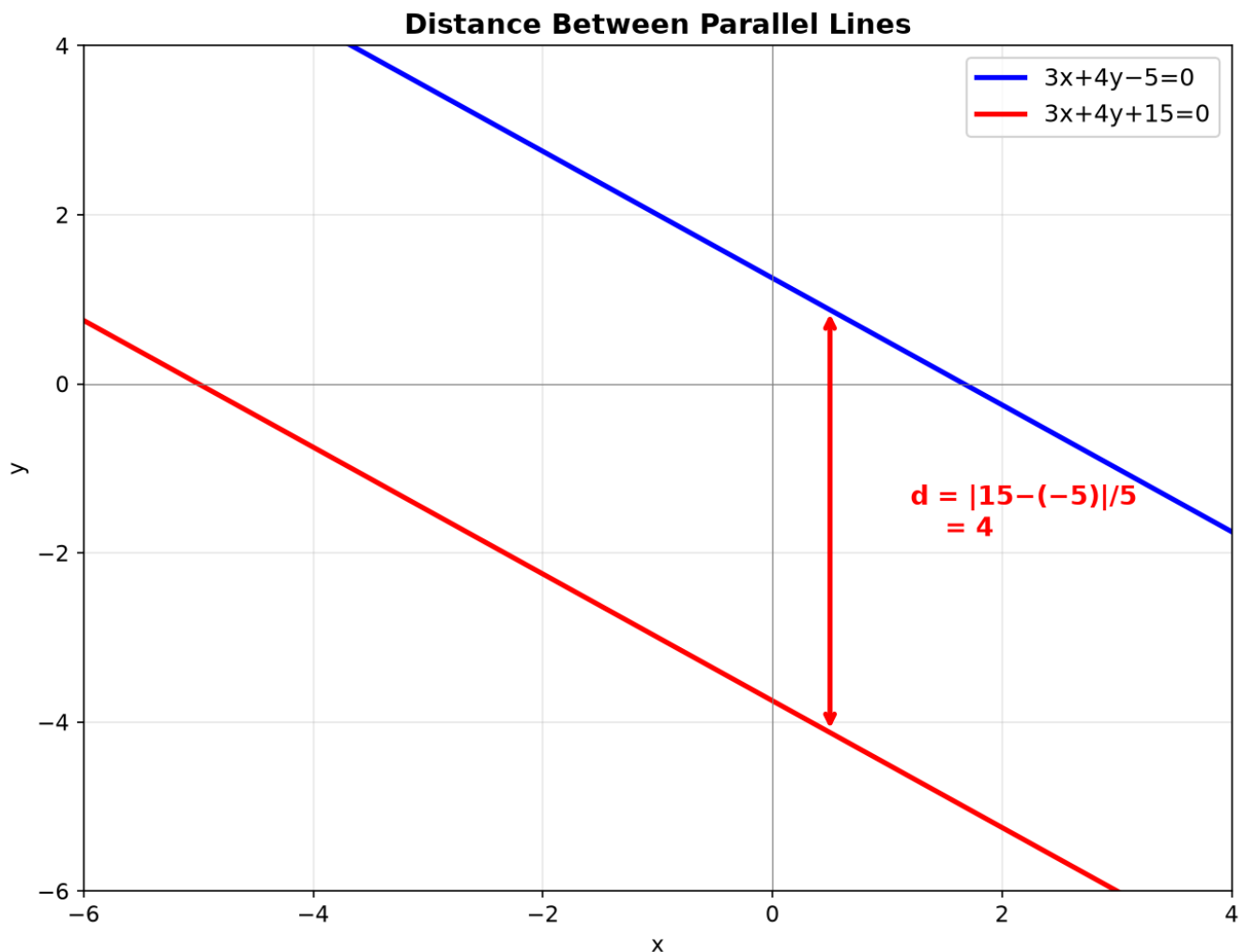
For parallel lines $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$ (same A, B — same normal, thus parallel):

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

$3x + 4y - 5 = 0$ and $3x + 4y + 15 = 0$:

$$d = \frac{|15 - (-5)|}{\sqrt{9+16}} = \frac{20}{5} = 4.$$

Why this works: Pick any point on the first line, compute its distance to the second. The result is independent of which point you pick — hence the clean formula.



Graph 9B-B2: Two parallel lines $3x + 4y - 5 = 0$ and $3x + 4y + 15 = 0$ are 4 units apart. The perpendicular segment (red dashed) has length $|15 - (-5)|/5 = 4$.

Example 9: Point-to-Circle Distance

Point $P(x_0, y_0)$ to circle center $C(h, k)$, radius R :

$$d = | \text{dist}(P, C) - R |$$

Case 1 — Point outside: $d = \text{dist}(P, C) - R$. The closest point lies on the line PC .

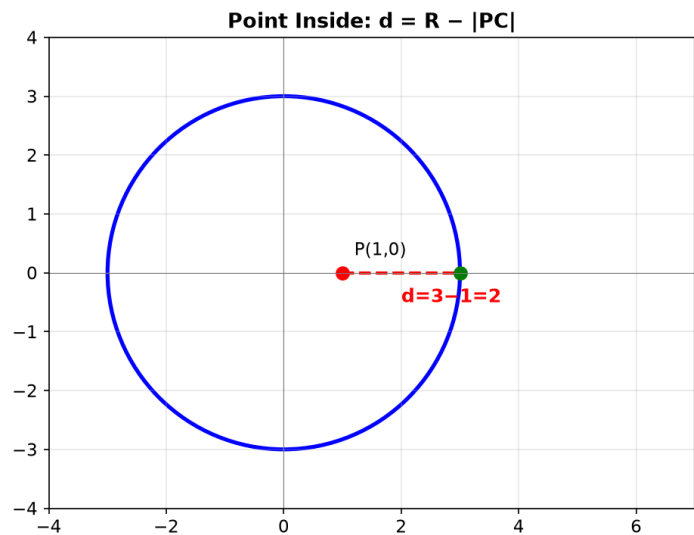
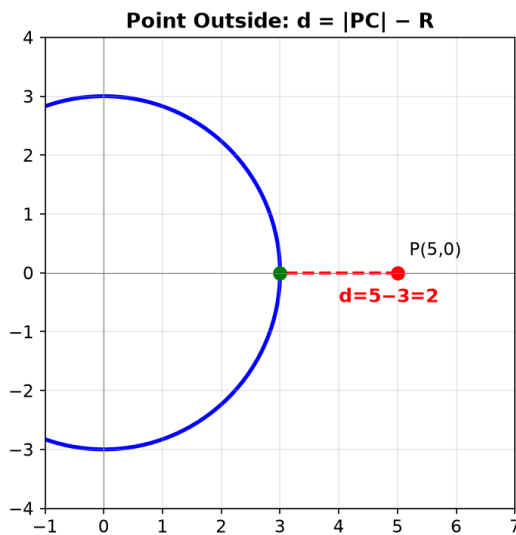
$(5, 0)$ to circle $x^2 + y^2 = 9$: $\text{dist}(P, C) = 5$, $R = 3$. $d = 5 - 3 = 2$.

Case 2 — Point inside: $d = R - \text{dist}(P, C)$.

$(1, 0)$ to circle $x^2 + y^2 = 9$: $\text{dist}(P, C) = 1$, $R = 3$. $d = 3 - 1 = 2$.

Case 3 — Point on the circle: $d = 0$.

Point-to-Circle Distance



Graph 9B-B3: Left — $P(5,0)$ outside the circle, distance = $5 - 3 = 2$. Right — $P(1,0)$ inside, distance = $3 - 1 = 2$. The shortest path always passes through the center.

Example 10: Tangent Lines to a Circle from an External Point

From point $P(x_0, y_0)$ outside circle $(x - h)^2 + (y - k)^2 = R^2$, two tangent lines can be

drawn.

Method:

- ① The tangent point T satisfies: $PT \perp CT$ (radius to tangent point is perpendicular to tangent).
- ② So $\triangle PTC$ is right-angled at T , with PC as hypotenuse.
- ③ $PT = \sqrt{PC^2 - R^2}$ (length of tangent segment).
- ④ Find T by solving $(x - h)^2 + (y - k)^2 = R^2$ and the perpendicular condition simultaneously.

Example: From $P(5, 0)$ to $x^2 + y^2 = 4$ (center $(0, 0)$, $R = 2$):

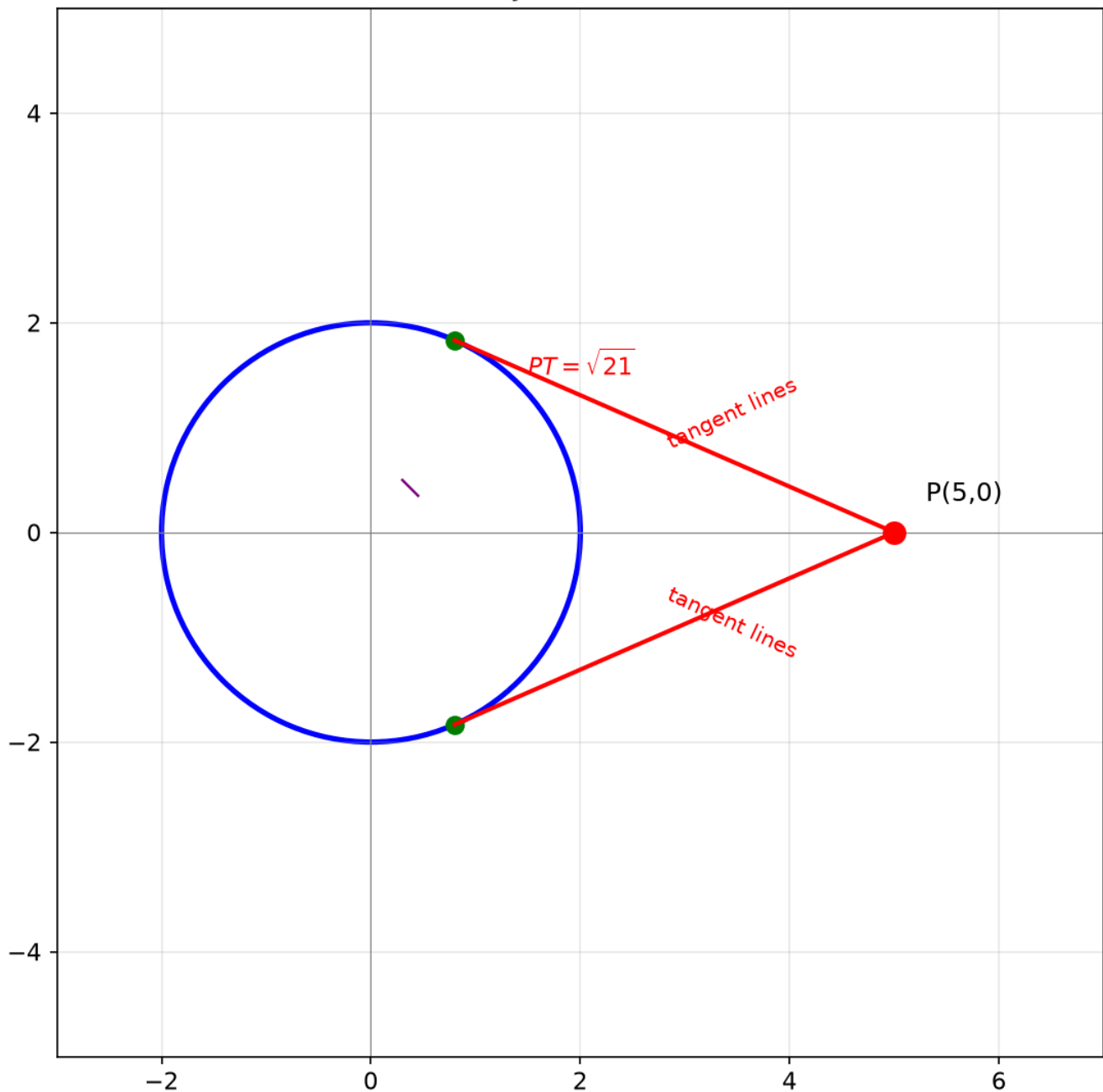
$$PC = 5, PT = \sqrt{25 - 4} = \sqrt{21}.$$

The tangent points satisfy $x^2 + y^2 = 4$ and the line from $(0, 0)$ to (x, y) being perpendicular to the line from $(5, 0)$ to (x, y) .

$$\text{Solving: } T = \left(\frac{4}{5}, \pm \frac{2\sqrt{21}}{5} \right).$$

Tangent Lines from External Point to Circle

$$x^2 + y^2 = 4, P(5,0)$$



Graph 9B-B4: From $P(5,0)$, two tangent lines touch the circle $x^2 + y^2 = 4$ at symmetric points. The tangent length is $\sqrt{21} \approx 4.58$. Right triangles at each tangent point.

Up to here (Part B) : Point-to-point (Pythagoras). Point-to-line (perpendicular projection).
Parallel lines distance. Point-to-circle. Tangent lines from external point.

Part C: Conic Sections — Deep Derivation

Cut a double cone with a plane at different angles. Four shapes emerge. Each has both an algebraic equation AND a geometric definition. Understanding both is the key.

Example 11: The Circle — Constant Distance from Center

Equation: $(x - h)^2 + (y - k)^2 = R^2$.

Geometric definition: All points exactly R units from center (h, k) .

General form: $x^2 + y^2 + Dx + Ey + F = 0$.

Complete the square in both x and y to recover center and radius.

Example: $x^2 + y^2 - 6x + 4y - 3 = 0$.

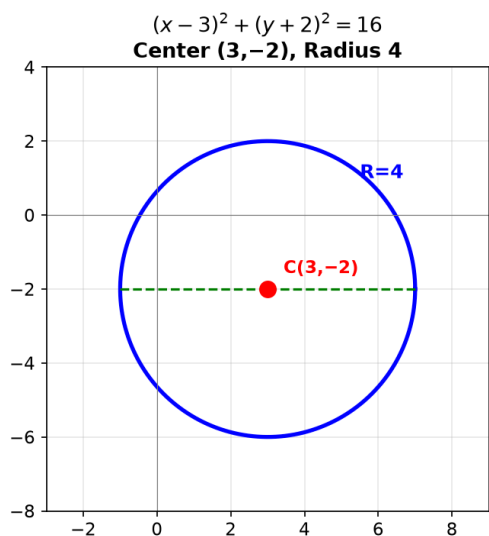
$$(x^2 - 6x) + (y^2 + 4y) = 3.$$

$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 3 + 9 + 4.$$

$$(x - 3)^2 + (y + 2)^2 = 16. \text{ Center } (3, -2), \text{ radius } 4.$$

Circle through three points: Solve a 3×3 system. If the points are (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , plug each into $x^2 + y^2 + Dx + Ey + F = 0$ and solve for D, E, F .

Circle — Standard Form and General Form



General Form:

$$x^2 + y^2 - 6x + 4y - 3 = 0$$

Step 1: Group

$$(x^2 - 6x) + (y^2 + 4y) = 3$$

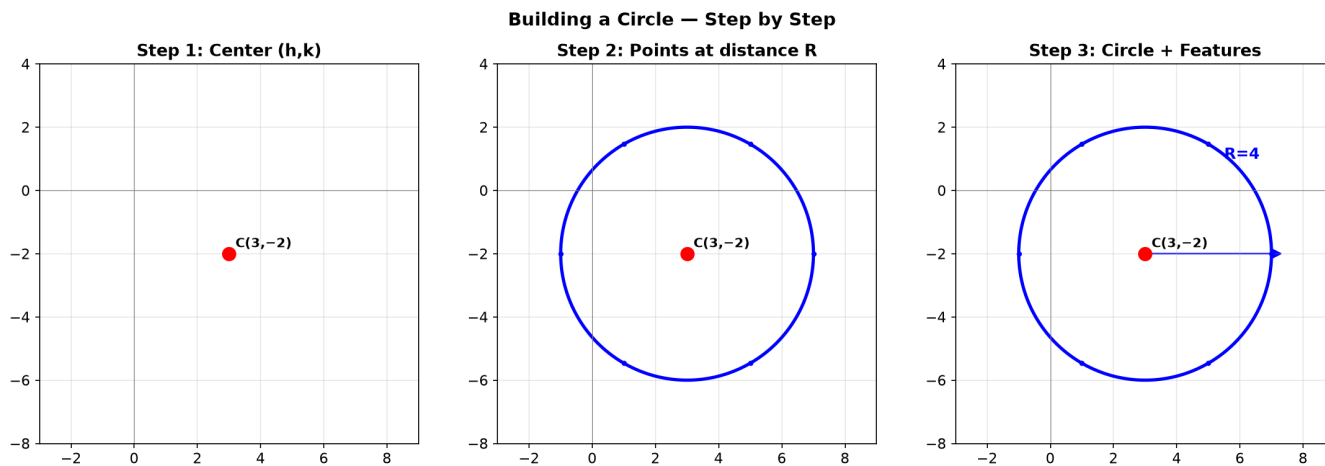
Step 2: Complete squares

$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 3 + 9 + 4$$

Step 3: Standard form

$$(x - 3)^2 + (y + 2)^2 = 16$$

Graph 9B-C1: Circle $(x - 3)^2 + (y + 2)^2 = 16$ with center $(3, -2)$ and radius 4. The general form $x^2 + y^2 - 6x + 4y - 3 = 0$ converts to standard form by completing the square in both x and y .



Graph 9B-S3: Building a circle in three steps. Step 1 — Mark center (h,k) . Step 2 — Draw all points at distance R . Step 3 — The circle with key features: center, radius, diameter, and the fact that x^2 and y^2 have equal coefficients.

Example 12: The Ellipse — Sum of Distances to Two Foci Is Constant

Equation: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, with $a \geq b > 0$.

Geometric definition: All points where $\text{dist}(P, F_1) + \text{dist}(P, F_2) = 2a$ (constant).

Key relationships (assuming $a \geq b$, major axis horizontal):

- Center: (h, k)
- Vertices: $(h \pm a, k)$ — endpoints of major axis
- Co-vertices: $(h, k \pm b)$ — endpoints of minor axis
- Foci: $(h \pm c, k)$ where $c^2 = a^2 - b^2$
- Eccentricity: $e = \frac{c}{a} = \sqrt{1 - \frac{b^2}{a^2}}$, $0 \leq e < 1$

Why $c^2 = a^2 - b^2$? For the rightmost vertex $(h + a, k)$:

Distance to right focus = $a - c$, to left focus = $a + c$. Sum = $(a - c) + (a + c) = 2a$. ✓

For the top co-vertex $(h, k + b)$:

Distance to right focus = $\sqrt{c^2 + b^2}$, same to left. Sum = $2\sqrt{c^2 + b^2}$.

Set equal to $2a$: $\sqrt{c^2 + b^2} = a \rightarrow c^2 + b^2 = a^2 \rightarrow c^2 = a^2 - b^2$.

Example: $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

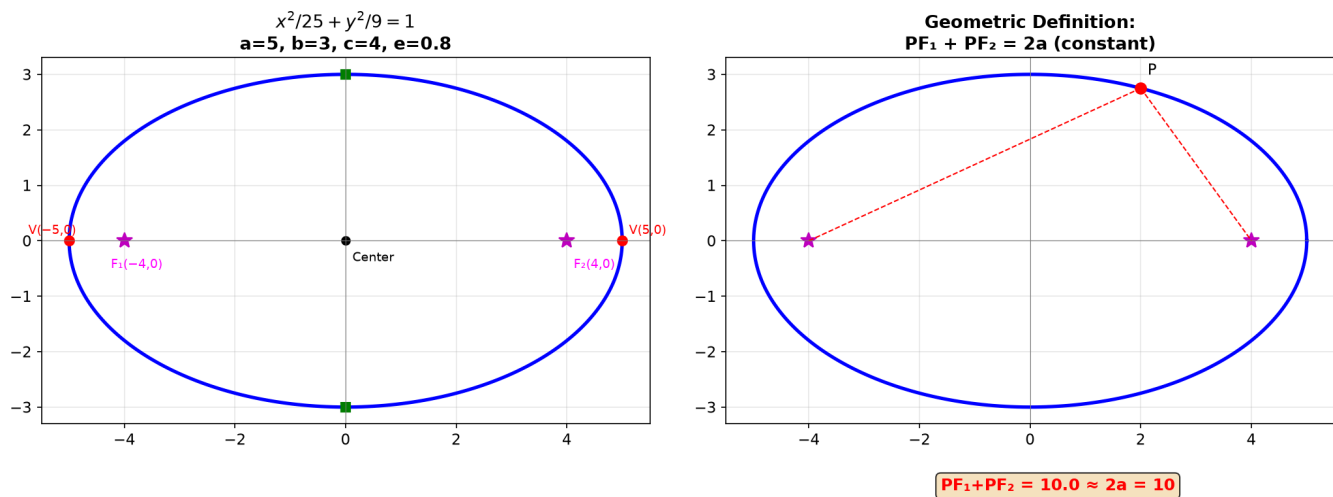
$a = 5, b = 3, c = \sqrt{25 - 9} = 4$.

Center $(0, 0)$, vertices $(\pm 5, 0)$, co-vertices $(0, \pm 3)$, foci $(\pm 4, 0)$.

Eccentricity $e = \frac{4}{5} = 0.8$.

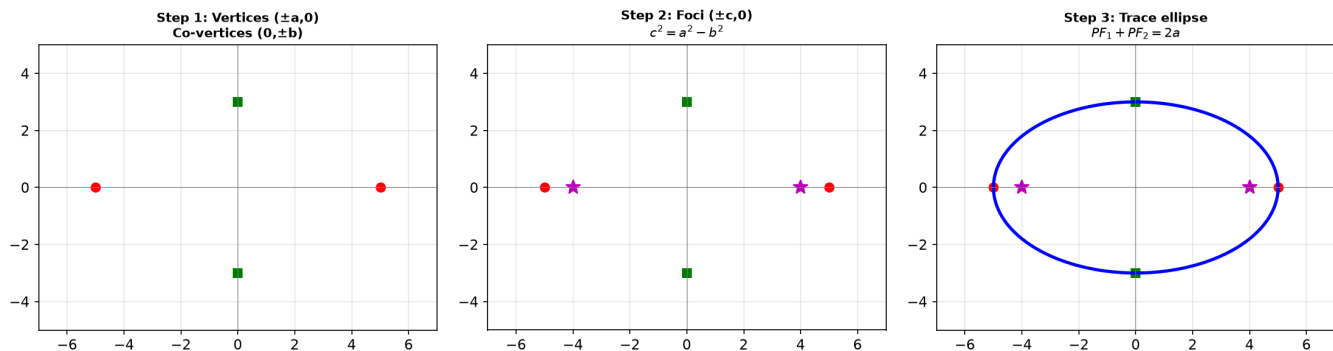
What if $b > a$? Then the major axis is vertical. Swap roles: $c^2 = b^2 - a^2$, foci at $(h, k \pm c)$.

Ellipse — Features and Geometric Definition



Graph 9B-C2: Ellipse $x^2/25 + y^2/9 = 1$. Left — All features labeled: center, vertices $(\pm 5, 0)$, co-vertices $(0, \pm 3)$, foci $(\pm 4, 0)$. Right — The constant-sum property: distance to F_1 + distance to $F_2 = 2a = 10$ for any point on the ellipse.

Building an Ellipse — Step by Step



Graph 9B-S4: Building an ellipse in three steps. Step 1 — Mark center, vertices $(\pm a)$, co-vertices $(\pm b)$. Step 2 — Mark foci at $(\pm c, 0)$ where $c^2 = a^2 - b^2$. Step 3 — Trace the ellipse: every point satisfies $PF_1 + PF_2 = 2a$.

Example 13: The Parabola — Equidistant from Focus and Directrix

Equation (vertical opening): $(x - h)^2 = 4p(y - k)$, or $y = \frac{1}{4p}(x - h)^2 + k$.

Geometric definition: Every point is equidistant from the **focus** and the **directrix**.

Key features (vertical parabola):

- Vertex: (h, k)
- Focus: $(h, k + p)$ — inside the "bowl"
- Directrix: $y = k - p$ — horizontal line behind the vertex
- $p > 0$: opens upward. $p < 0$: opens downward.
- $|4p|$ is the **focal width** (latus rectum) — the width at the focus.

Deriving p from $y = ax^2 + bx + c$:

Complete the square to vertex form $y = a(x - h)^2 + k$.

Then $a = \frac{1}{4p}$, so $p = \frac{1}{4a}$.

Example: $y = \frac{1}{2}x^2 - 2x + 3$.

① $y = \frac{1}{2}(x^2 - 4x) + 3 = \frac{1}{2}(x - 2)^2 - 2 + 3 = \frac{1}{2}(x - 2)^2 + 1$.

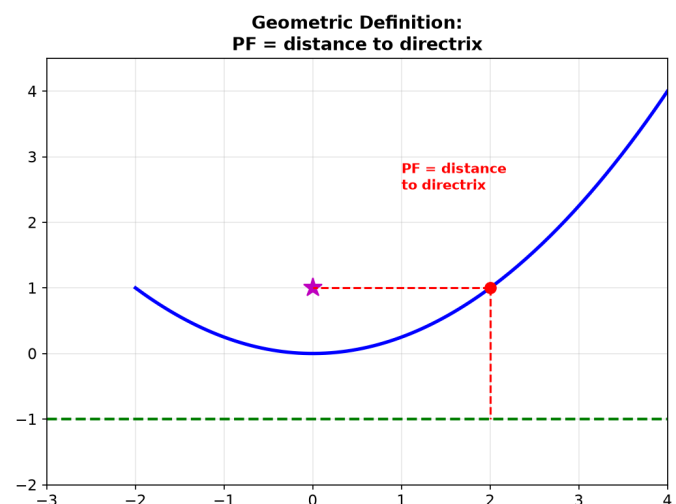
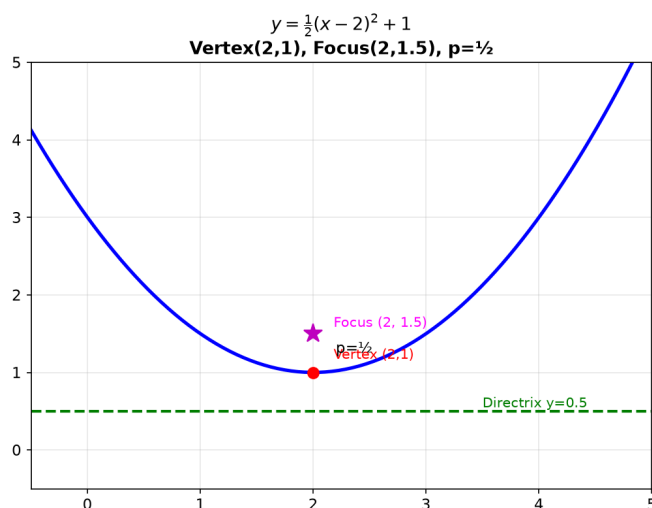
② $a = \frac{1}{2}$, so $p = \frac{1}{4(1/2)} = \frac{1}{2}$.

③ Vertex $(2, 1)$, focus $(2, 1.5)$, directrix $y = 0.5$.

Horizontal parabola: $(y - k)^2 = 4p(x - h)$. Opens right if $p > 0$, left if $p < 0$.

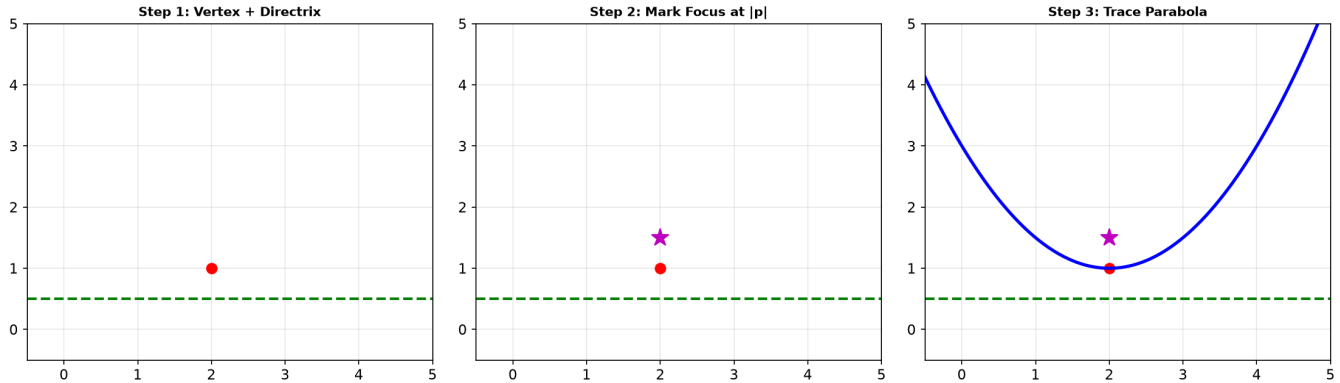
Focus $(h + p, k)$, directrix $x = h - p$.

Parabola — Focus, Directrix, and Definition



Graph 9B-C3: Parabola $y = (x - 2)^2 / 2 + 1$. Left — All features: vertex $(2, 1)$, focus $(2, 1.5)$, directrix $y = 0.5$, focal width $|4p| = 2$. Right — The equidistance property: for any point on the parabola, distance to focus = distance to directrix.

Building a Parabola — Step by Step



Graph 9B-S5: Building a parabola in three steps. Step 1 — Mark vertex (h, k) and directrix. Step 2 — Mark focus at distance $|p|$ from vertex. Step 3 — Trace the parabola: every point is equidistant from focus and directrix.

Example 14: The Hyperbola — Absolute Difference of Distances to Two Foci Is Constant

Equation (horizontal opening): $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$

Geometric definition: All points where $|\text{dist}(P, F_1) - \text{dist}(P, F_2)| = 2a$ (constant).

Key features (horizontal hyperbola):

- Center: (h, k)
- Vertices: $(h \pm a, k)$ — where the curve crosses the transverse axis
- Foci: $(h \pm c, k)$ where $c^2 = a^2 + b^2$
- Asymptotes: $y - k = \pm \frac{b}{a}(x - h)$ — the lines the branches hug at infinity
- Eccentricity: $e = \frac{c}{a} = \sqrt{1 + \frac{b^2}{a^2}}, e > 1$

Why $c^2 = a^2 + b^2$? For the right vertex $(h + a, k)$:

Distance to right focus = $|a - c|$, to left focus = $a + c$.

Difference = $(a + c) - |a - c|$. Since $c > a$: $|a - c| = c - a$, so difference = $(a + c) - (c - a) = 2a$. ✓

Asymptote derivation: As $x, y \rightarrow \infty$, the 1 on the RHS becomes negligible:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} \approx 0 \rightarrow \frac{y^2}{b^2} \approx \frac{x^2}{a^2} \rightarrow y \approx \pm \frac{b}{a}x.$$

Example: $\frac{x^2}{9} - \frac{y^2}{4} = 1.$

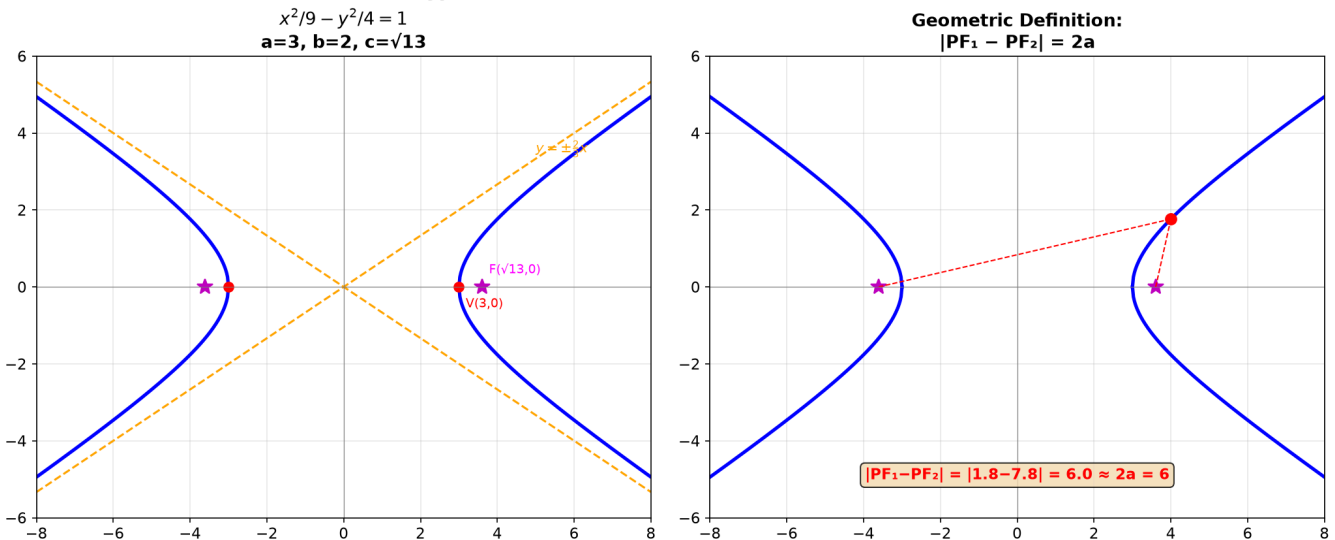
$a = 3, b = 2, c = \sqrt{9 + 4} = \sqrt{13} \approx 3.61.$

Center $(0, 0)$, vertices $(\pm 3, 0)$, foci $(\pm \sqrt{13}, 0)$.

Asymptotes: $y = \pm \frac{2}{3}x.$

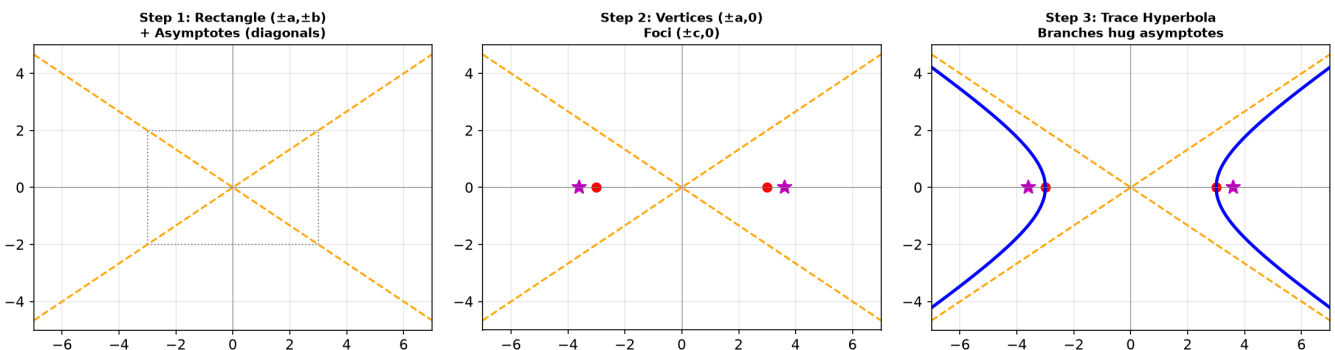
Vertical opening: $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1.$ Vertices at $(h, k \pm a)$, asymptotes $y - k = \pm \frac{a}{b}(x - h).$

Hyperbola — Features and Geometric Definition



Graph 9B-C4: Hyperbola $x^2/9 - y^2/4 = 1$. Left — All features: center, vertices $(\pm 3, 0)$, foci $(\pm \sqrt{13}, 0)$, asymptotes $y = \pm \frac{2}{3}x$. Right — The constant-difference property: $|PF_1 - PF_2| = 2a = 6$.

Building a Hyperbola — Step by Step



Graph 9B-S6: Building a hyperbola in three steps. Step 1 — Draw the fundamental rectangle $(\pm a, \pm b)$ and its diagonals (asymptotes). Step 2 — Mark vertices $(\pm a, 0)$ and foci $(\pm c, 0)$. Step 3

— Trace both branches hugging the asymptotes.

Example 15: Identifying Conics — The Discriminant Method

General second-degree equation: $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

Discriminant: $\Delta = B^2 - 4AC$.

Δ	Type	Special case
$\Delta < 0$	Ellipse	$A = C$ and $B = 0 \rightarrow$ Circle
$\Delta = 0$	Parabola	
$\Delta > 0$	Hyperbola	

How to remember: Same sign pattern as the discriminant of a quadratic ($b^2 - 4ac$). Ellipse = no xy crossing (negative), parabola = tangent (zero), hyperbola = crossing (positive).

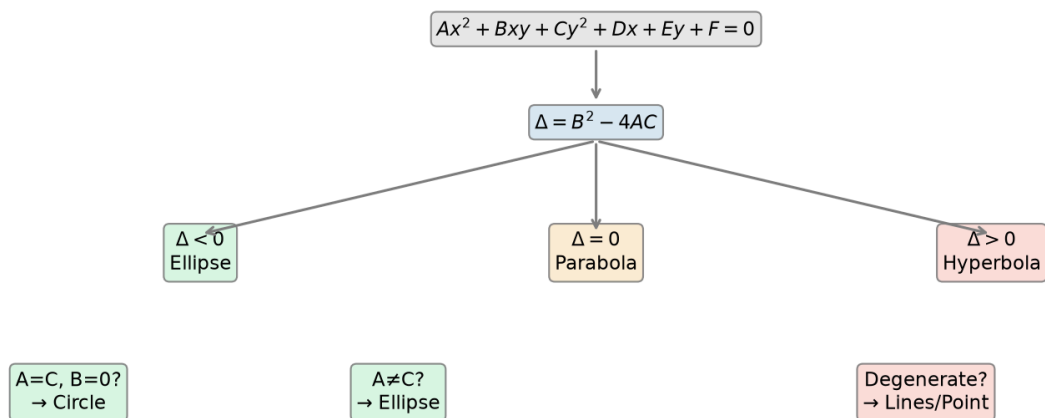
Examples:

- $x^2 + 4y^2 - 6x + 8y + 9 = 0$: $B = 0$, $A = 1$, $C = 4$. $\Delta = 0 - 16 = -16 < 0 \rightarrow$ **Ellipse**.
- $x^2 - y^2 = 4$: $B = 0$, $A = 1$, $C = -1$. $\Delta = 0 - 4(-1) = 4 > 0 \rightarrow$ **Hyperbola**.
- $y = x^2$: Rewrite as $x^2 - y = 0$. $B = 0$, $A = 1$, $C = 0$. $\Delta = 0 \rightarrow$ **Parabola**.

Degenerate cases: If the equation can't be satisfied by real points.

- $x^2 + y^2 = -1 \rightarrow$ no real points (empty set).
- $x^2 - y^2 = 0 \rightarrow (x - y)(x + y) = 0 \rightarrow$ two intersecting lines.
- $x^2 = 0 \rightarrow$ a single line (y -axis, counted twice).

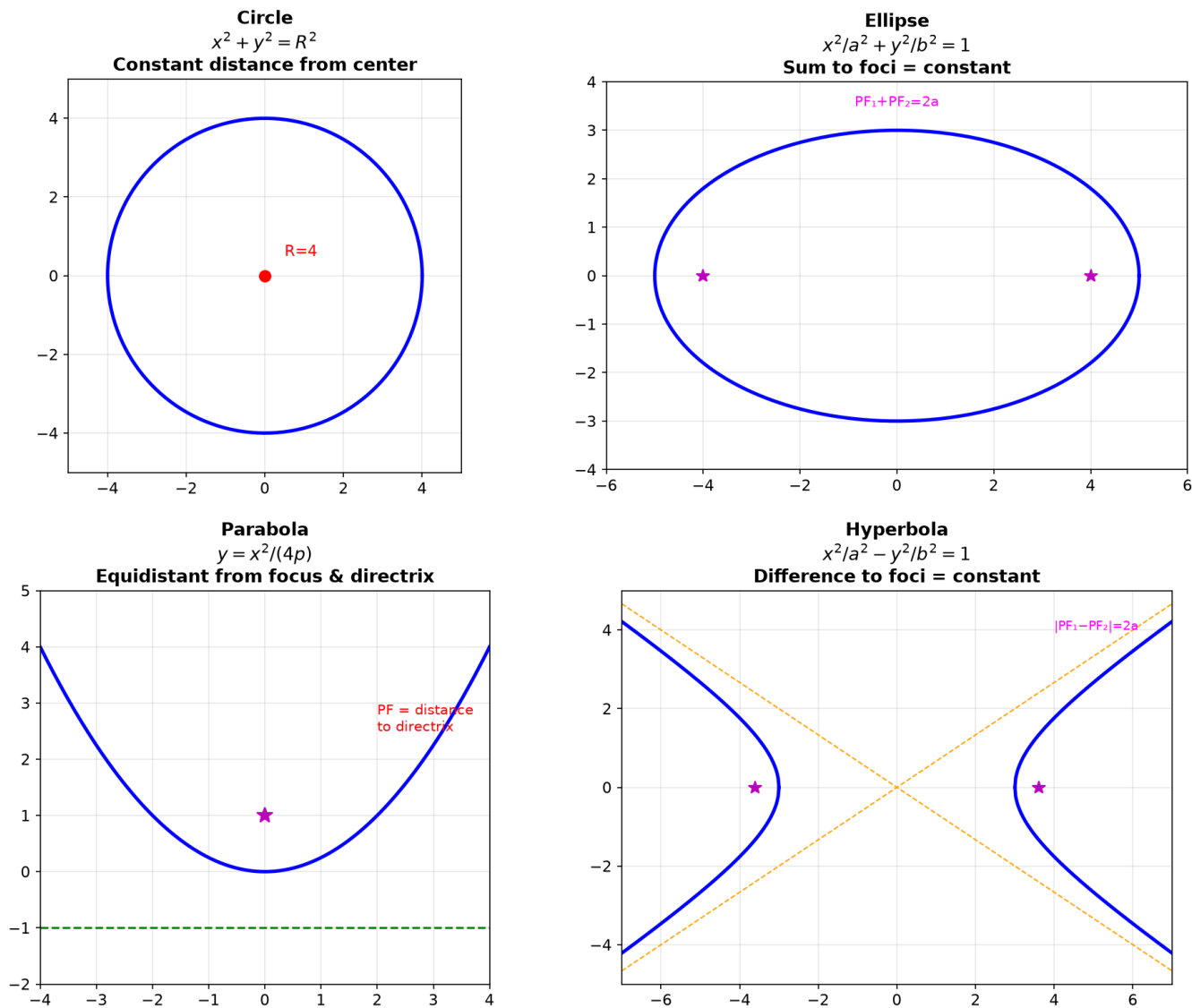
Conic Identification — The Discriminant Method



Complete the square to find center/vertex.
Degenerate: $x^2 + y^2 = -1$ (empty), $x^2 - y^2 = 0$ (two lines).

Graph 9B-C5: Identifying conics by discriminant $B^2 - 4AC$. Negative $\rightarrow$ ellipse (or circle). Zero $\rightarrow$ parabola. Positive $\rightarrow$ hyperbola. Complete the square to find the center/vertex and specific features.

Four Conic Sections — Side by Side



Graph 9B-C6: All four conic sections compared — Circle ($x^2 + y^2 = R^2$), Ellipse ($x^2/a^2 + y^2/b^2 = 1$), Parabola ($y = x^2/(4p)$), Hyperbola ($x^2/a^2 - y^2/b^2 = 1$). Each with its geometric definition labeled.

Up to here (Part C) : Circle (R constant). Ellipse (sum to foci $= 2a$, $c^2 = a^2 - b^2$). Parabola (focus = directrix, $p = 1/(4a)$). Hyperbola (difference to foci $= 2a$, $c^2 = a^2 + b^2$). Identify by discriminant $B^2 - 4AC$.

Part D: Parametric Curves — Motion Described by a Parameter

A parametric curve is $(x(t), y(t))$ — position as a function of time. The parameter t animates the shape.

Example 16: Parametric Line and Line Segment

Line through (x_1, y_1) with direction (a, b) :

$$(x(t), y(t)) = (x_1 + at, y_1 + bt), t \in \mathbb{R}.$$

Line segment from (x_1, y_1) to (x_2, y_2) :

$$(x(t), y(t)) = (x_1 + (x_2 - x_1)t, y_1 + (y_2 - y_1)t), t \in [0, 1].$$

At $t = 0$: starting point. At $t = 0.5$: midpoint. At $t = 1$: ending point.

Eliminating the parameter: Solve for t from one equation, substitute into the other.

From $x = x_1 + at$, $t = \frac{x-x_1}{a}$, then $y = y_1 + b\frac{x-x_1}{a} = y_1 + \frac{b}{a}(x - x_1)$.

This is point-slope form — slope $m = b/a$.

Example 17: Parametric Circle and Ellipse

Circle of radius R centered at (h, k) :

$$(x(t), y(t)) = (h + R \cos t, k + R \sin t), t \in [0, 2\pi].$$

Ellipse centered at (h, k) :

$$(x(t), y(t)) = (h + a \cos t, k + b \sin t), t \in [0, 2\pi].$$

Eliminating the parameter: $\cos^2 t + \sin^2 t = 1$.

$$\left(\frac{x-h}{a}\right)^2 + \left(\frac{y-k}{b}\right)^2 = 1 \text{ — the ellipse equation.}$$

Direction: As t increases from 0, the point moves **counterclockwise** starting from the rightmost

point $(h + a, k)$.

Example 18: The Cycloid — A Wheel's Trail

A point on the rim of a wheel of radius R rolling along the x -axis:

$$(x(t), y(t)) = (R(t - \sin t), R(1 - \cos t)), t \geq 0.$$

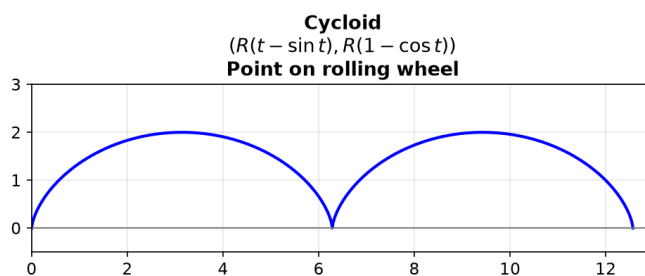
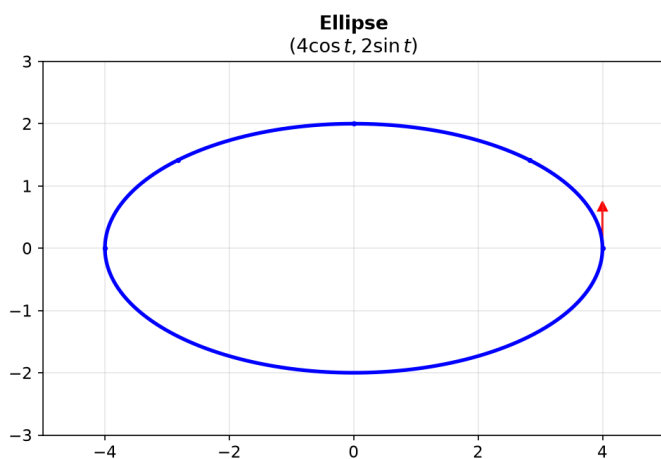
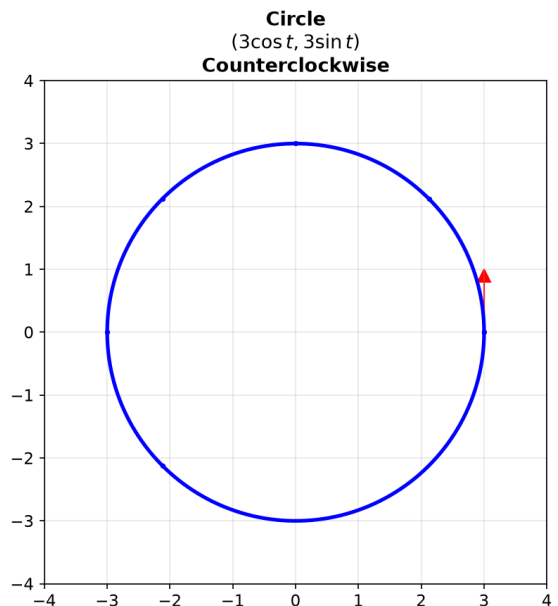
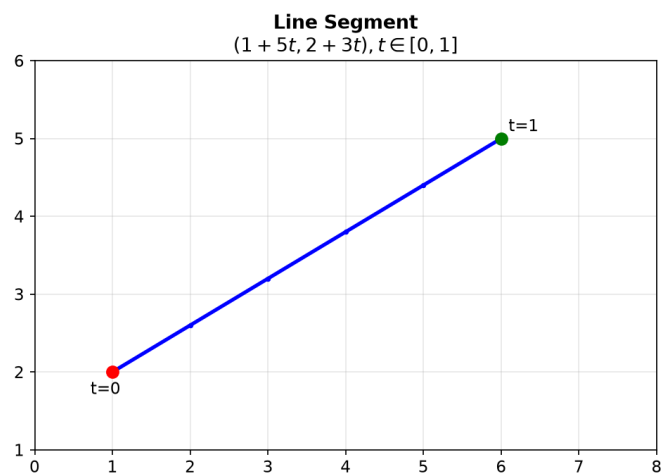
At $t = 0$: $(0, 0)$ — the point touches the ground.

At $t = \pi$: $(\pi R, 2R)$ — the highest point.

At $t = 2\pi$: $(2\pi R, 0)$ — back to the ground, one full arch.

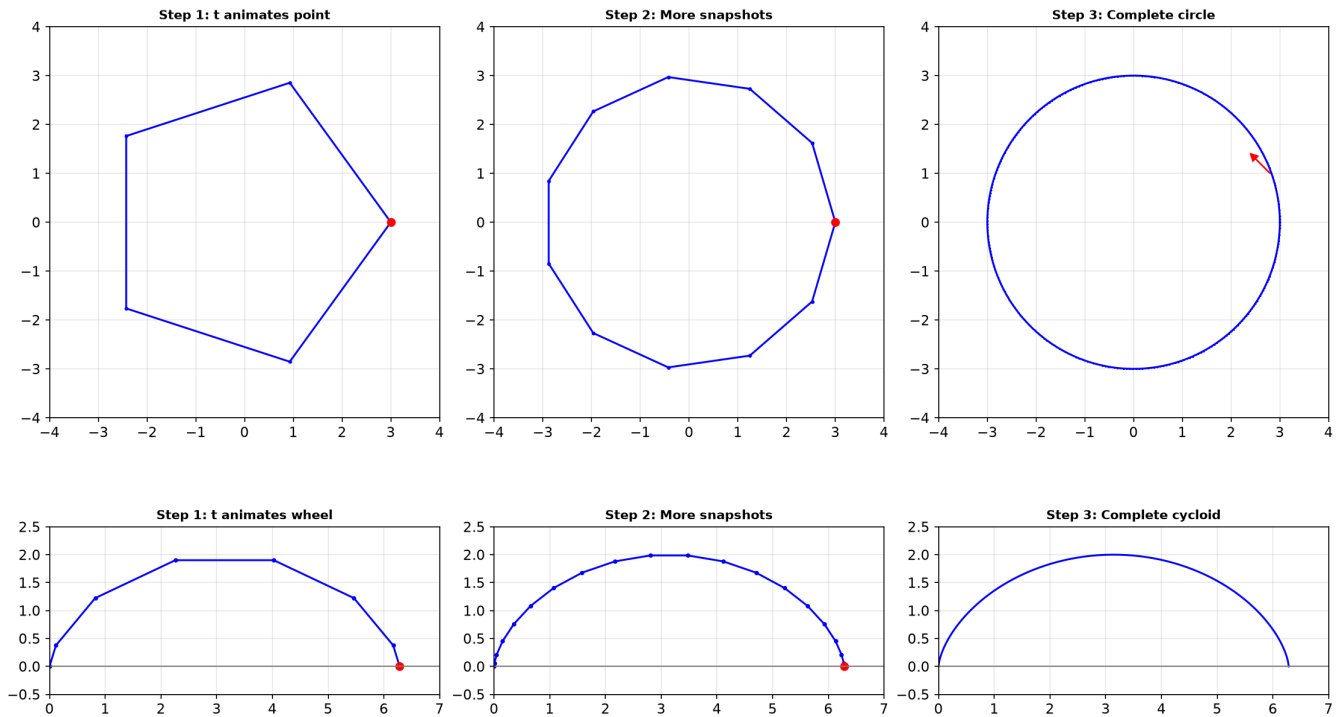
Arch length (one full rotation): $t \in [0, 2\pi]$, horizontal displacement = $2\pi R$.

Parametric Curves



Graph 9B-D1: Four parametric curves. Top-left — Line segment: uniform motion from A to B. Top-right — Circle: constant speed counterclockwise. Bottom-left — Ellipse: stretched circle. Bottom-right — Cycloid: a point on a rolling wheel traces arches.

Building Parametric Curves — Step by Step



Graph 9B-S7: Parametric curves built in three stages each. Left column — The parameter t animates a point along the curve. Middle — Multiple snapshots show the motion. Right — The complete curve with direction arrows.

Up to here (Part D) : Parametric line/segment, circle, ellipse, cycloid. Eliminate t to recover Cartesian form. Parameter direction indicates motion.

Part E: Area and Polygon Geometry

Closed shapes have area. Triangles from coordinates, polygons by decomposition — pure algebra.

Example 19: Triangle Area from Coordinates

Given vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) :

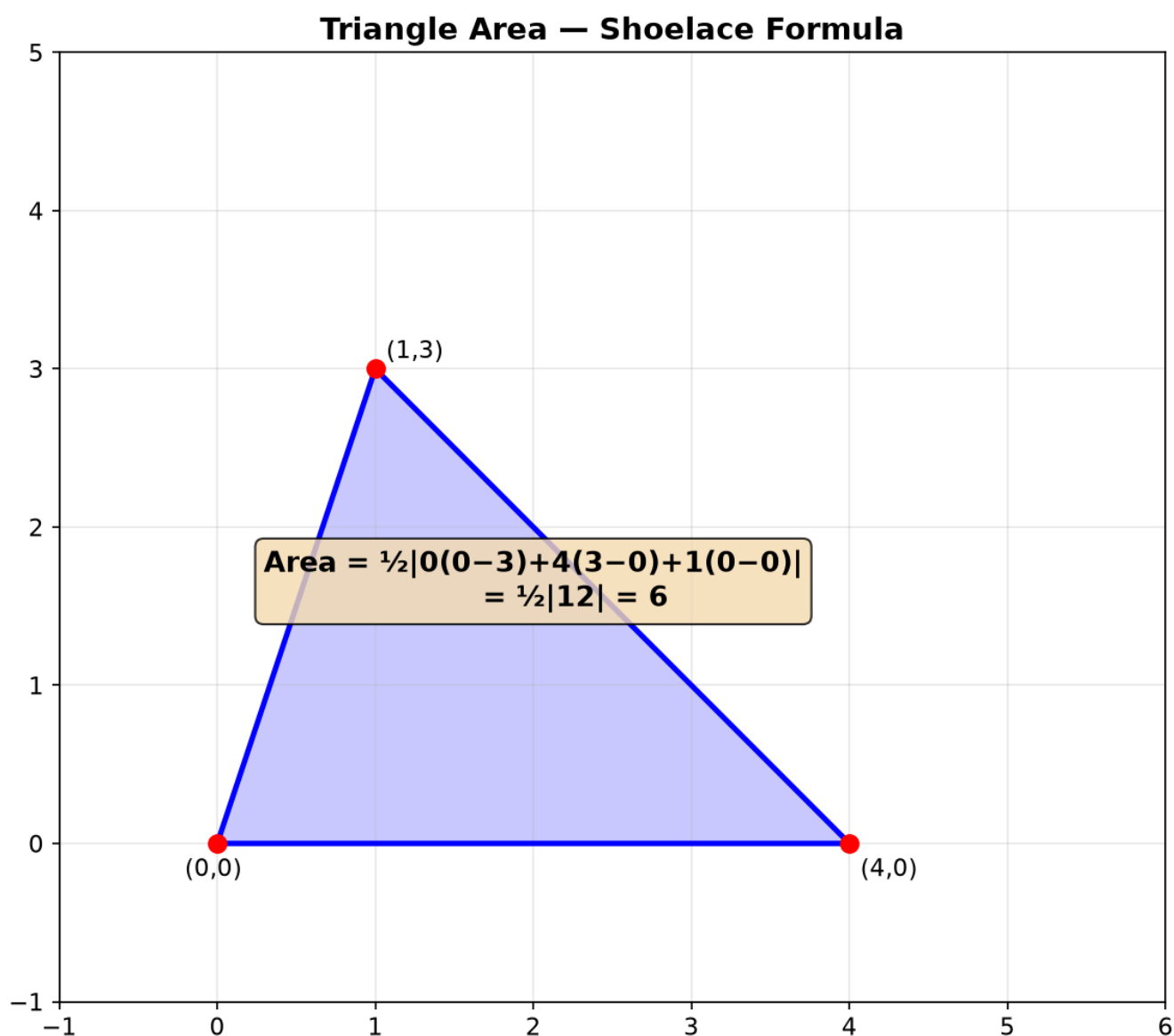
$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

This is the **shoelace formula** (or surveyor's formula) for a triangle.

$(0, 0)$, $(4, 0)$, $(1, 3)$:

$$\text{Area} = \frac{1}{2} |0(0 - 3) + 4(3 - 0) + 1(0 - 0)| = \frac{1}{2} |12| = 6.$$

Geometric interpretation: Arrange coordinates in a column, repeat the first, multiply diagonally (down then up), take half the absolute difference.



Graph 9B-E1: Triangle with vertices $(0,0)$, $(4,0)$, $(1,3)$. Area = 6 by the shoelace formula. The same result comes from $\frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot 4 \cdot 3 = 6$.

Example 20: Polygon Area — The Shoelace Formula

For any simple polygon with vertices (x_1, y_1) , (x_2, y_2) , $\dots$, (x_n, y_n) in order:

$$\text{Area} = \frac{1}{2} \left| \sum_{i=1}^n x_i y_{i+1} - \sum_{i=1}^n x_{i+1} y_i \right|$$

where $(x_{n+1}, y_{n+1}) = (x_1, y_1)$ (close the loop).

Example: Quadrilateral $(0, 0)$, $(5, 0)$, $(4, 3)$, $(1, 4)$.

Arrange vertically:

$$(0, 0) \rightarrow 0 \cdot 0 - 0 \cdot 5 = 0$$

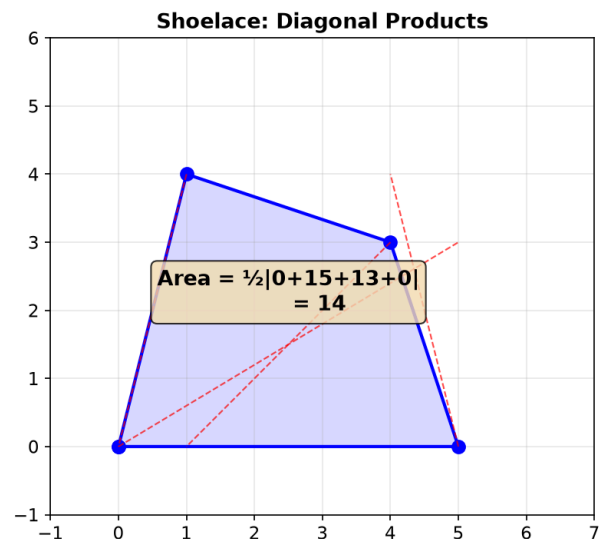
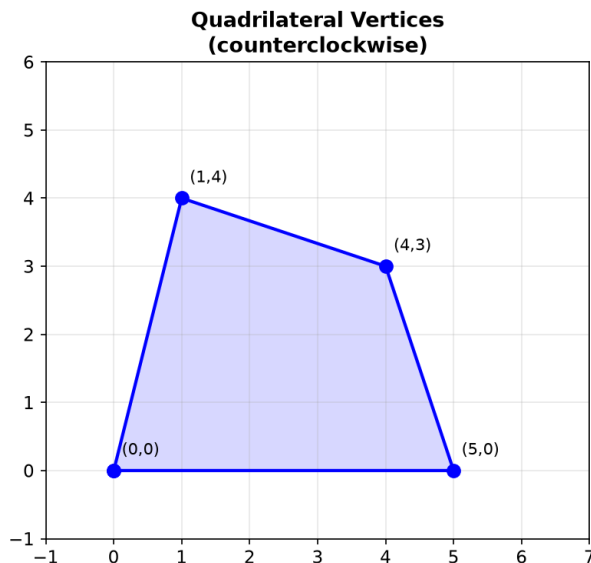
$$(5, 0) \rightarrow 5 \cdot 3 - 4 \cdot 0 = 15$$

$$(4, 3) \rightarrow 4 \cdot 4 - 1 \cdot 3 = 13$$

$$(1, 4) \rightarrow 1 \cdot 0 - 0 \cdot 4 = 0$$

$$\text{Sum} = 28. \text{Area} = \frac{1}{2} |28| = 14.$$

Polygon Area — The Shoelace Formula



Graph 9B-E2: The shoelace formula applied to a quadrilateral. Left — Vertices in counterclockwise order. Middle — The diagonal products. Right — Area = 14. This works for any simple polygon.

Example 21: Point Reflection — Mirroring Across a Line

Given point $P(x_0, y_0)$ and line $Ax + By + C = 0$, the reflected point $P'(x', y')$:

- ① The midpoint of PP' lies on the line.
- ② PP' is perpendicular to the line (direction (A, B)).
- ③ $P' = P - 2 \cdot \frac{Ax_0 + By_0 + C}{A^2 + B^2} \cdot (A, B)$.

Example: Reflect $(1, 5)$ across $x + y = 0$ (the line $y = -x$).

$$A = 1, B = 1, C = 0.$$

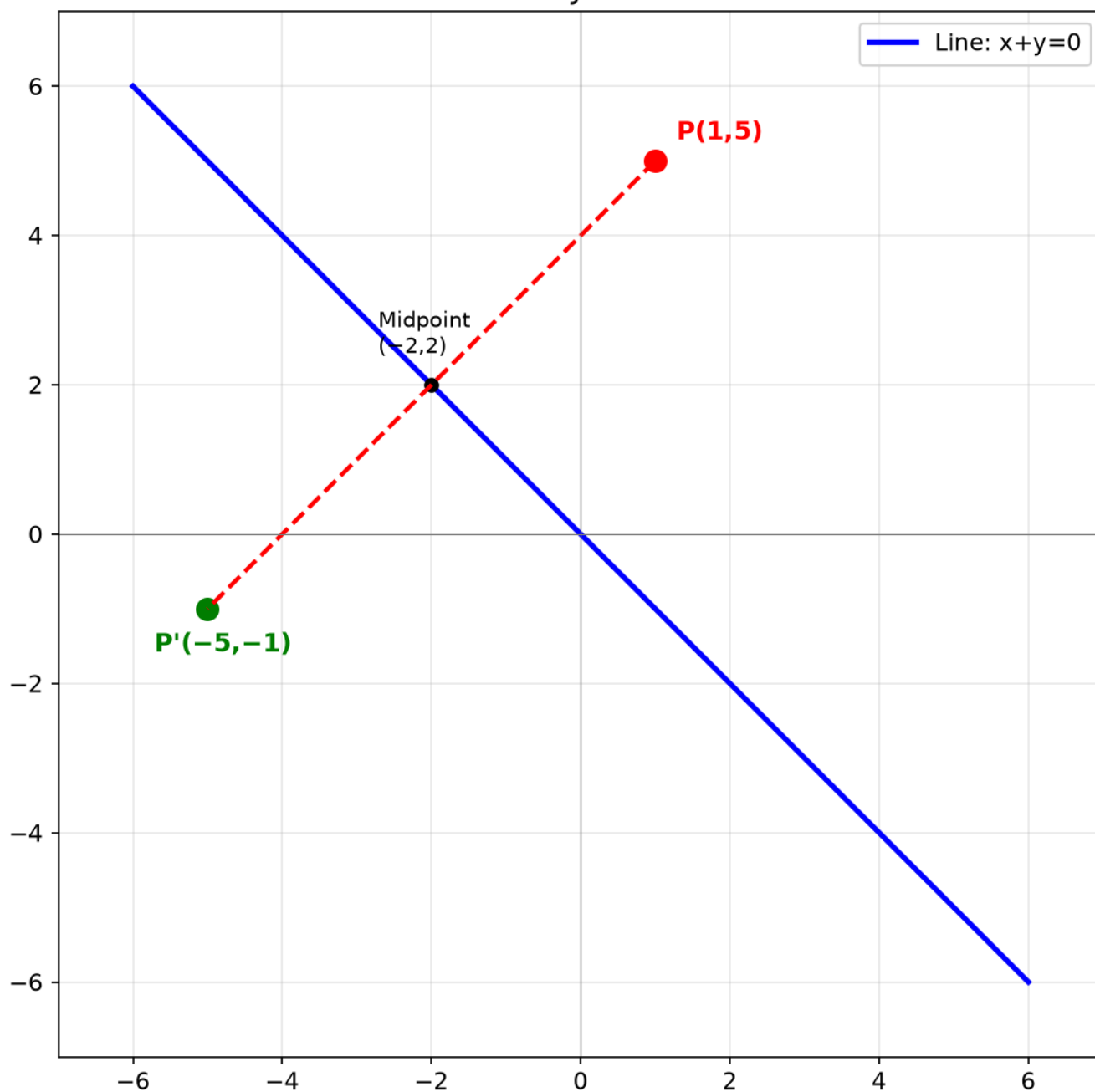
$$Ax_0 + By_0 + C = 1 + 5 = 6, A^2 + B^2 = 2.$$

$$P' = (1, 5) - 2 \cdot \frac{6}{2} \cdot (1, 1) = (1, 5) - (6, 6) = (-5, -1).$$

Check: midpoint $(-2, 2)$ lies on $y = -x$. ✓

Point Reflection Across a Line

$$x + y = 0$$



Graph 9B-E3: Reflecting point $(1,5)$ across the line $x+y=0$ gives $(-5,-1)$. The line is the perpendicular bisector of the segment connecting point and its reflection. Midpoint $(-2,2)$ lies on the line.

Common Mistakes

Mistake 1: Forgetting to take absolute value in distance formulas

Wrong: $d = \frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}}$. **Right:** $d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$. Distance is always non-negative.

Mistake 2: Confusing a and b roles in ellipse

Wrong: " a is always the x -denominator." **Right:** a is the semi-major axis (larger denominator). If $b > a$, the major axis is vertical, and $c^2 = b^2 - a^2$.

Mistake 3: Forgetting the minus sign in hyperbola asymptotes

Wrong: Asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm \frac{a}{b}x$.

Right: $y = \pm \frac{b}{a}x$. The slope is the ratio of the y -coefficient's square root to the x -coefficient's square root.

Mistake 4: Using $c^2 = a^2 - b^2$ for hyperbola

Wrong: Same c formula for both ellipse and hyperbola. **Right:** Ellipse: $c^2 = a^2 - b^2$. Hyperbola: $c^2 = a^2 + b^2$. The sign flips because the equation changes from $+$ to $-$.

Mistake 5: Treating discriminant $\Delta = 0$ as "no conic"

Wrong: "If $\Delta = 0$, there's no conic." **Right:** $\Delta = 0$ means **parabola** — the plane is parallel to the side of the cone.

What We Just Did

- (1) The Line: 5 forms. Slope = $\tan(\theta)$. Parallel ($m_1=m_2$), perpendicular ($m_1m_2=-1$).
Angle between lines: $\tan(\phi) = |(m_2-m_1)/(1+m_1m_2)|$.
Midpoint and section formula.
- (2) Distance: point-to-point (Pythagoras), point-to-line (perpendicular projection), parallel lines, point-to-circle, tangent lines from external point.
- (3) Conic Sections – geometric definitions + algebraic equations:
Circle: $(x-h)^2+(y-k)^2 = R^2$. R constant.
Ellipse: $PF_1+PF_2 = 2a$. $c^2 = a^2-b^2$. $0 \leq e < 1$.
Parabola: $PF = \text{distance to directrix}$. $p = 1/(4a)$. $e = 1$.
Hyperbola: $|PF_1-PF_2| = 2a$. $c^2 = a^2+b^2$. $e > 1$.
Identify by $\Delta = B^2-4AC$.
- (4) Parametric curves: $(x(t), y(t))$ – line, circle, ellipse, cycloid.
- (5) Area: triangle (shoelace), polygon (extended shoelace). Point reflection.

Practice 1

A line passes through $(1, 4)$ and $(5, -2)$. Write it in all five forms. What is its x -intercept?

→ Reference: **Examples 1, 2**

Solutions: [Solutions](#)

Practice 2

Find the acute angle between the lines $y = 3x + 1$ and $2x + y = 5$.

→ Reference: **Example 4**

Solutions: [Solutions](#)

Practice 3

Find the distance from $(2, -1)$ to the line $4x - 3y + 5 = 0$, and find the foot of the perpendicular.

→ Reference: **Example 7**

Solutions: [Solutions](#)

Practice 4

Find the center, vertices, foci, and eccentricity of $9x^2 + 25y^2 = 225$. Then draw the ellipse.

→ Reference: **Example 12**

Solutions: [Solutions](#)

Practice 5

A parabola has focus $(2, 1)$ and directrix $y = -3$. Find its equation in standard form and vertex form.

→ Reference: **Example 13**

Solutions: [Solutions](#)

Practice 6: Real Battle

A hyperbola has asymptotes $y = \pm \frac{4}{3}x$ and passes through $(5, 0)$. Find its equation, foci, and eccentricity. Then find the distance from the origin to either asymptote (they are the same).

→ Reference: **Examples 7, 14**

Solutions: [Solutions](#)

Basic Algebra Drill — 2D Geometry (15 Problems)

Pure computation + 5 geometry-insight problems (marked ♦).

- D1.** Write the equation of the line through $(3, -1)$ with slope $\frac{2}{5}$ in point-slope, slope-intercept, and general form.
- D2.** Find the midpoint of the segment joining $(-3, 7)$ and $(9, -5)$.
- D3.** Determine whether the lines $6x - 3y + 1 = 0$ and $y = 2x - 4$ are parallel, perpendicular, or neither.
- D4.** Find the distance from $(-2, 8)$ to the line $5x + 12y - 10 = 0$.
- D5.** Find the center and radius of $x^2 + y^2 + 8x - 6y + 21 = 0$.
- D6.** Identify the conic: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. Find its center and vertices.
- D7.** Identify the conic: $y^2 - 4x^2 = 16$. Find its asymptotes and foci.
- D8.** Find the equation of the parabola with vertex $(3, -2)$ and focus $(3, 1)$.
- D9.** A line segment from $(2, 3)$ to $(8, 9)$ is divided by a point that is twice as far from the first endpoint as from the second. Find the coordinates of the dividing point.
- D10.** Find the tangent length from $(13, 0)$ to the circle $x^2 + y^2 = 25$. Find the equation of one tangent line.
- ♦ **D11.** Without computing the distance formula explicitly, explain geometrically why $(3, 4)$ to the

line $3x + 4y = 0$ is exactly 5 units away. (Hint: draw the right triangle.)

◆ **D12.** Two points $A(0, 0)$ and $B(6, 0)$ form the base of a triangle. The third vertex C moves such that the area is always 12. What curve does C trace? Describe it.

◆ **D13.** A line with slope m passes through $(0, 0)$. For what values of m does the line intersect the circle $x^2 + y^2 - 4x - 4y + 4 = 0$ at exactly one point? (Geometric interpretation: these are the tangent lines from the origin.)

◆ **D14.** The parametric curve $(x(t), y(t)) = (t^2, t)$ for $t \in \mathbb{R}$ describes a parabola. Find its Cartesian equation and identify the vertex and focus. (Hint: $x = t^2 = (y)^2$.)

◆ **D15.** Consider the family of lines $y = mx + (1 - m)$ for all real m . Show that ALL these lines pass through a single fixed point. Find that point.

Solutions: [Solutions](#)

Advanced Algebra Drill — 2D Geometry (15 Problems)

Multi-step geometric reasoning + 5 geometry-insight problems (marked ◆).

A1. Find the equation of the circle passing through the three points $(1, 2)$, $(4, 1)$, and $(2, -3)$.

A2. Find the distance between the parallel lines $3x - 4y + 7 = 0$ and $6x - 8y - 15 = 0$.

A3. A line through $(2, 3)$ has slope m . It intersects the circle $x^2 + y^2 = 20$ at two points. For what values of m is the line tangent (only one intersection point)?

A4. The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has eccentricity $\frac{3}{5}$ and passes through $(4, \frac{12}{5})$. Find a and b .

A5. Find the equation of the hyperbola with foci $(\pm 5, 0)$ that passes through $(4, 0)$.

A6. Reflect the point $(3, -2)$ across the line $y = 2x + 1$. Find the coordinates of the reflected point.

A7. The parametric curve $(x(t), y(t)) = (\sec t, \tan t)$ for $t \in (-\pi/2, \pi/2)$ describes part

of a hyperbola. Eliminate t to find the Cartesian equation (hint: $\sec^2 t - \tan^2 t = 1$).

A8. Find the area of the quadrilateral with vertices $(1, 1)$, $(6, 2)$, $(5, 7)$, $(2, 5)$.

A9. A line passes through $(3, 0)$ and intersects the parabola $y = x^2$ at two points P and Q . Find the equation of the line if the midpoint of PQ lies on the line $x = 1$.

A10. Two circles: $x^2 + y^2 = 4$ and $(x - 3)^2 + (y - 4)^2 = 1$. Find the shortest distance between them. Do they intersect?

◆ **A11.** A point moves such that its distance from $(1, 0)$ is always twice its distance from the line $x = 4$. Find the equation of its path and identify the conic. What is its eccentricity?

◆ **A12.** The line $y = mx + c$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Derive the condition on m and c . Use this to show that $c^2 = a^2m^2 + b^2$.

◆ **A13.** Four points A, B, C, D have coordinates $(0, 0)$, $(a, 0)$, (a, b) , $(0, b)$ — a rectangle. Now rotate the rectangle by angle θ about the origin. Find the area of the rotated rectangle using the shoelace formula, and verify that area is invariant under rotation.

◆ **A14.** Two perpendicular lines through the origin intersect the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at four points. Show that the sum $\frac{1}{OP_1^2} + \frac{1}{OP_2^2}$ is constant for any choice of perpendicular lines, where OP_1 and OP_2 are distances from the origin to the intersection points.

◆ **A15.** Consider all chords of the parabola $y = x^2$ that pass through the point $(0, 1)$. Prove that the midpoints of these chords all lie on another parabola. Find its equation.

Solutions: [Solutions](#)

Today's Procedure

Step 1: Lines – Know all 5 forms and convert fluently.

Parallel/perpendicular by slope. Midpoint and section formula.

Step 2: Distance – Point-to-line formula (derive it once, use it forever).

Point-to-circle = |distance to center – R|.

Step 3: Conic Sections – For each one, know BOTH the algebraic equation

AND the geometric definition (sum/difference/distance relationships).

Identify by discriminant $\Delta = B^2 - 4AC$.

Step 4: Parametric curves – $(x(t), y(t))$ describes motion.

Eliminate t to get back to Cartesian form.

Step 5: Area – Shoelace formula for any polygon.

Point reflection = perpendicular bisector method.

How to Read These Symbols

Symbol	Reads as	Meaning
$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	"distance equals square root of delta-x squared plus delta-y squared"	2D distance (Pythagorean theorem)
$m = \frac{y_2 - y_1}{x_2 - x_1}$	"m equals delta y over delta x" / "slope"	slope = rise over run
$y = mx + b$	"y equals m x plus b" / "slope-intercept form"	line: m = slope, b = y-intercept
$y - y_1 = m(x - x_1)$	"y minus y1 equals m times x minus x1"	point-slope form
$Ax + By + C = 0$	"A x plus B y plus C equals zero"	general form,

Symbol	Reads as	Meaning
		normal vector (A,B)
$m_1 m_2 = -1$	"m1 times m2 equals negative one"	perpendicular lines condition
$\tan \phi = \left \frac{m_2 - m_1}{1 + m_1 m_2} \right $	"tan phi equals absolute value of..."	angle between two lines
$(x - h)^2 + (y - k)^2 = R^2$	"x minus h squared plus y minus k squared equals R squared"	circle: center (h,k), radius R
$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	"ellipse equation"	ellipse: a = semi-major, b = semi-minor
$(x - h)^2 = 4p(y - k)$	"x minus h squared equals 4p times y minus k"	parabola: vertex (h,k), focal length p
$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	"hyperbola equation"	hyperbola: opens left-right
$e = \frac{c}{a}$	"e equals c over a"	eccentricity
$(x(t), y(t))$	"x of t, y of t"	parametric curve with parameter t
$\frac{1}{2} x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) $	"one half times absolute value of..."	triangle area by coordinates (shoelace)

Terminology

What we call it	Math term	Notation
slope	slope / gradient	m

What we call it	Math term	Notation
y-intercept	y-intercept	b
rise over run	slope	$\Delta y / \Delta x$
normal vector	normal vector	(A, B) perpendicular to $Ax + By + C = 0$
midpoint	midpoint	$M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
section formula	internal division	divides segment in ratio $m : n$
perpendicular bisector	perpendicular bisector	line perpendicular to segment through its midpoint
circle/ellipse/ parabola/hyperbola	conic sections	from $Ax^2 + Bxy + Cy^2 + \dots = 0$
focus, directrix	focus, directrix	geometric definition of parabola
foci (plural of focus)	foci	geometric definition of ellipse, hyperbola
eccentricity	eccentricity	e — measures "flatness": 0=circle, <1=ellipse, =1=parabola, >1=hyperbola
latus rectum	latus rectum / focal width	chord through focus perpendicular to axis, length $ 4p $
parametric curve	parametric equations	$(x(t), y(t))$
shoelace formula	surveyor's formula	polygon area from vertex coordinates
degenerate conic	degenerate conic	equation reduces to point, line(s), or empty set